3.2.1 Molecules of Elements

A molecule of an element consists of fixed number of one types of atom chemically combined. Table 3.1 shows that gases are made up of two atoms of the same element.

3.2.2 Molecules of Compounds

Molecule of a compound consists of a fixed number of different types of atoms chemically combined. For example, let us look at the model of a water molecule below. Each molecule of water consists of one oxygen atom and two hydrogen atoms. The ratio of oxygen and hydrogen atoms remains fixed whether water is in liquid, solid or gaseous state. This principle applies to the molecules of all compounds. Compounds with different atoms are given in Table 3.2.



Model of water molecule



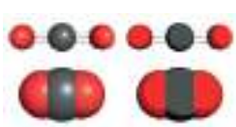
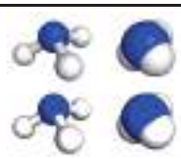
Bismuth in diarrhea medicine

Bismuth is an element that occurs naturally. It is combined with other elements to make medicine for treating diarrhea.

Table 3.1 Compounds with same atoms

Molecule	Chlorine Gas	Oxygen Gas	Nitrogen Gas
Molecule Diagram	Cl Cl	O O	N N
Molecule Model (Ball-and-Stick)	 Chlorine Molecule	 Oxygen Molecule	 Nitrogen Molecule

Table 3.2 Compounds with different atoms

Molecule	Carbon dioxide	Ammonia	Hydrogen Chloride
Molecule Diagram	O C O	H N H H	H Cl
Molecule Model (Ball-and-Stick)	 Carbon-dioxide Molecule	 Ammonia Molecule	 Hydrogen Chloride

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